sometimes you can compensate for systematic accuracy errors through calibration.

- *Environmental tolerance* refers to the constraints under which your sensor operates. Industrial environments might expose sensors to extreme temperatures, vibration, dust, or corrosive chemicals. Outdoor sensors need weatherproofing. Some environments require intrinsically safe designs to prevent explosion risks, while others demand non-contaminating sensors that won't compromise sample purity in pharmaceutical, food, or clean room applications.
- *Power consumption* is the primary constraint that can shape your entire sensing strategy. Wired power is not always available, forcing you to trade off between power and functionality. When considering battery power for sensors, you will need to carefully consider factors such as sampling rate and the cost of wireless communication to minimize the operational costs of replacing batteries across your sensor fleet.

3.2 Selecting the right sensors for your digital twin

Choosing the right sensor involves more than matching measurement type to physical parameter. A temperature sensor that works perfectly in a home environment might fail in an industrial setting, while a highly accurate sensor becomes useless if it can't communicate reliably with your digital twin platform. Deciding what to measure, as well as how and when to measure it, must be driven by the objectives you have defined based on the decision-making improvements you aim to achieve. Here are factors to consider when selecting sensors for your digital twin.

3.2.1 Define what you're measuring

Working backwards from the objectives of your digital twin, start by identifying the specific physical parameter that best indicates the condition you're trying to monitor. This isn't always obvious, for example, equipment overheating might manifest as elevated temperature, increased vibration, or changing electrical current draw.

Once you've identified the right parameter, establish the expected operating range. For my home's irrigation system, soil moisture typically varies